THE OPTION TO QUIT: THE EFFECT OF EMPLOYEE STOCK OPTIONS ON TURNOVER[1]

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Abstract

We show that in the years following a large broad-based employee stock option (BBSO) grant, employee turnover falls at the granting firm. We find evidence consistent with a causal relation by exploiting unexpected changes in the value of unvested options. A large fraction of the reduction in turnover appears to be temporary with turnover increasing in the 3rd year following the year of the adoption of the BBSO plan. We also find that the effect of BBSO plans is larger at market leaders, identified as firms with high industry-adjusted market-to-book ratios, market share or industry-adjusted profit margins, as measured at the time of the grant.

Keyword*:* Employee stock options, turnover

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When firms grant broad-based employee stock options (BBSOs), they provide an incentive for employees to remain with the firm until those options vest.[2] This suggests that BBSO grants should be associated with turnover reductions at granting firms. It is unclear, however, whether we would expect to see an effect in practice. First, it is common for competitors to poach employees with options by granting a signing bonus to "make the employees whole". In this case, BBSOs impose costs on competitors, but do not affect turnover. Second, as options eventually vest, a reduction in turnover may be temporary, as employees merely delay joining competitors until after their options vest. We exploit a large panel of establishment-level data to show that turnover is reduced following the granting of BBSOs, and that the relationship appears causal.

We document that this reduction in turnover is temporary, to a great extent. After the options vest, we observe an increase in turnover nearly comparable in magnitude to the initial reduction. As such, it appears that BBSO grants are primarily delaying, as opposed to preventing, turnover. The increase in turnover once the options vest suggests that the mechanism through which BBSO grants reduce turnover is associated with anticipation of a deferred compensation award as opposed to feelings of greater loyalty towards the firm following the granting of stock-based compensation.

We also find that BBSO plans are associated with stronger reduction in turnover in up-markets, when growth options are high and retaining key employees is likely to be especially valuable. We also find that the reduction in turnover is larger at market-leading firms. Firms that

[2] A BBSO plan is a grant of stock options to a broad set of a firm's employees, rather than solely to executives. As with executive options, strike prices are typically at par when the options are granted, and options vest over time. The average vesting period is three to four years (Crimmel and Schildkraut, 2001, Oyer and Schaefer, 2005), but options may not expire for several years after vesting. The 2000 National Compensation Survey found that options expire 8.9 years after they are granted, on average. A more detailed review of the institutional details and usage of BBSO plans can be found in Aldatmaz and Ouimet (2011).

have a high industry-adjusted market-to-book ratio, high market share, or high industry-adjusted profit margin at the time of the BBSO grant benefit the most from the issuance of employee stock options, suggesting complementarity between option grants and market leadership.

Turnover is important. The direct costs of turnover are high: Hansen (1997) estimates that the cost of hiring and training a new worker equals 150% to 175% of her annual pay. The indirect costs are high as well: Brachet and David (2011) find that turnover accounts for twice as much organizational forgetting as the more traditionally understood skill depreciation. Consistent with these facts, firms are placing an increasing emphasis on reducing turnover.[3]

Theory suggests that BBSOs are a natural solution. Typically, these options vest in 3 years (Crimmel and Schildkraut, 2001, Oyer and Schaefer, 2005), and an employee who departs before her options have vested will forfeit their value. Furthermore, as Oyer (2004) shows, insofar as her outside employment opportunities are likely to increase when stock prices are higher, the incentives provided by BBSOs against quitting are likely to be stronger when quitting is more appealing.

We connect quarterly, establishment-level Census data on turnover with ExecuComp data on stock option grants to estimate the effect of implementing a broad based stock option plan on employee turnover. In the years following a large BBSO grant, turnover falls at the granting firm. We document evidence consistent with causality by exploiting exogenous variation in the value of unvested BBSO grants. If the existence of unvested options reduces turnover, then the effect should be larger when the firm's stock price rises following the option grant, thereby increasing the value of the unvested options. While we may observe this effect in the data, it is not sufficient evidence that the BBSO grant is causing reduced turnover. If a firm's stock price is

[3]This emphasis on turnover is evidenced by the increasing frequency of managerial contracts providing a bonus if turnover is kept below a specified level. See, for example:
http://online.wsj.com/article/SB121477771122614255.html, accessed on November 7, 2012.

predictable for the manager establishing the BBSO plan, then a variety of omitted variables could alternatively explain such results.

Thus, we instead focus on market returns which occur after the option grant. To the extent that own-firm returns are correlated with market returns, a firm's option values will increase when market returns are high. Moreover, market-wide returns are presumably unpredictable for the manager. This effect should be stronger at firms whose stock price is highly correlated with their peers and, indeed, we show that this is the case.

While the primary goal of the paper is to document the effect of BBSO plans on turnover, we have two sets of additional results of interest to economists. First, we confirm results consistent with the theory of Oyer (2004) that BBSOs are especially valuable in reducing turnover because the "carrot" they provide adjusts to the temptation to quit that employees face. We show that BBSOs, on average, lead to relatively greater turnover reductions following positive market returns. This is important because the loss of valuable employees is presumably more costly following positive market returns.[4]

Second, we find that turnover is more affected by BBSOs when the firm is a market leader or has greater growth opportunities. This may be surprising, because more successful firms have lower turnover without BBSOs, leaving less scope for turnover reductions. These results suggest complementarity between explicit and implicit retention incentives, and support Core and Guay (2001)'s finding that firms with greater growth options are more likely to implement BBSO plans.

Prior work has suggested that BBSOs can improve firm performance by better aligning employee incentives, conserving cash, screening employees, and reducing turnover. Core and

[4] Hiring is more difficult when labor markets are tight, suggesting that direct turnover costs are likely higher in good times.

Guay (2001) find evidence that firms with higher growth opportunities, high capital needs, and lower access to capital markets use BBSO plans more. Based on this evidence, they conclude that aligning incentives, conserving cash and retaining employees drive the implementation of BBSO plans. Ittner, Lambert, and Larcker (2003) study new economy firms and argue that BBSO plans are used as a substitute for more expensive means of monitoring employees. Carter and Lynch (2004) show that firms that re-price options subsequently experience lower turnover, as compared to a matched set of firms, where forfeited in-the-money options are used as a proxy for turnover. Oyer and Schaefer (2005) calibrate an agency model and conclude that screening employees and reducing turnover can justify the implementation of BBSO plans. Bergman and Jenter (2007) find evidence that BBSO plans are most common among firms with superior prior stock performance, supporting their theory that BBSO plans might be justified as a means to screen employees based on their sentiment about the company's prospects. Kedia and Rajgopal (2009) find evidence that the implementation of BBSO plans might be motivated by the granting behavior of peer firms. Blasi, Freeman, Mackin, and Kruse (2010) find that workers owning stock options report that they are less likely to look for a new job. Lastly, Hochberg and Lindsey (2010) explore the relationship between BBSO plans and subsequent firm operating performance. Instrumenting for the decision to adopt a BBSO plan with peer firms' BBSO implementation, they find that BBSO plans lead to superior operating performance.

While reducing turnover has been suggested as one motivation, among many others, in some of these studies, none of them has tested the claim directly. In this paper, we provide evidence that there is indeed a negative relationship between BBSO plans and employee turnover. We document evidence suggesting this relationship is causal, thereby supporting the claim that retaining employees is one of the reasons why companies implement BBSO plans.

Section I presents a description of the data used in our analyses and some summary statistics. Section II includes our baseline results and some analyses of the effect of BBSO plans on turnover in up versus down markets. In Section III, we analyze the effectiveness of BBSOs in reducing turnover as we vary firm attributes and Section IV concludes. Definitions of dependent and independent variables in our analyses can be found in Table 1.

I Data

We use four primary data sources in the analysis. We use two databases from the US Census Bureau. The Longitudinal Employer-Household Dynamics (LEHD) database contains information on hiring and employment at U.S. establishments and the Longitudinal Business Database (LBD) contains information about establishment characteristics, including ultimate owner. We identify BBSO plans through Standard and Poor's ExecuComp database. Compustat provides information on firm characteristics. We discuss each in turn.

A. Longitudinal Employer-Household Dynamics Data

The LEHD program has created a unique panel database matching employees to employers and tracking the relationship over time. The data are primarily derived from state unemployment records and contain information on employee age, gender, wages, hiring, and departure dates.[5] The data also include the employer's geographic location and industry. The data cover all establishments within a state, once the state is covered by the LEHD program. An

[5] See Abowd et al (2009) for more detail.

establishment[6] is identified as any separate geographic address at which at least one employee works. The panel begins in 1992 for several states and the coverage of states increases over time.

We use the LEHD to estimate employee turnover. Employee turnover includes all voluntary and involuntary departures from a firm, and is estimated as:

$$T_{i,j,t} = \left(H_{i,j,t} - \max\left(\left(E_{i,j,t} - E_{i,j,t-1}\right), 0\right)\right) / E_{i,j,t-1} \qquad (1)$$

where *T* is turnover, *H* is the number of new hires and *E* is the level of employment, at firm *i*, establishment *j*, and at the beginning of quarter *t*. Turnover covers both full-time and part-time workers. We limit our sample to full quarter employees, defined as workers employed over the previous quarter, as these employees are most likely to be granted stock options. Turnover is estimated at the establishment-level and is measured quarterly. The LEHD reports the hiring, separations and earnings of workers in each quarter by age and gender.

It is worth discussing two simple cases to illustrate how turnover is defined. In the first case, suppose the firm does not change the number of employees at an establishment in quarter t, but one worker quits and is replaced. Then $E_{i,j,t} - E_{i,j,t-1} = 0$ and $H_{i,j,t} = 1$, so $T_{i,j,t} = 1/E_{i,j,t-1}$. As should be intuitive, a fraction 1/*E* of the workforce turns over in this case. In the second case, the firm hires one more person at the establishment without anybody departing. Then $E_{i,j,t} - E_{i,j,t-1} = 1$ and $H_{i,j,t} = 1$, so $T_{i,j,t} = \frac{1-1}{E_{i,j,t-1}} = 0$. This firm has expanded, not experienced turnover.

[6] A business establishment is part of a firm defined by having a particular geographic location. For example, a law firm with an office in Boston and an office in New York would have two business establishments. Likewise, a manufacturing firm with three different plants operating in different locations, e.g., two in Illinois and one in Wisconsin, would have three business establishments.

Wage data in the LEHD are available for all employees at a given establishment as well as in breakdowns by age and gender cohorts. The information included in employee earnings includes base salary as well as any other payment taxed as ordinary income, such as bonuses.

B. Longitudinal Business Database

The LBD is a panel data set that tracks all U.S. business establishments with at least one employee or positive payroll. The database is formed by linking years of the standard statistical establishment list (SSEL) maintained by the Internal Revenue Service of the U.S. Treasury Department. The SSEL contains the federal employer identification number (EIN), physical address, and name of each business establishment that files tax returns in the U.S.

The LBD contains information on the number of employees working for an establishment and total annual establishment payroll.[7] The LBD allows us to identify establishment age, based on when the establishment enters the database.[8] The LBD also identifies the owner of all establishments. Finally, we use the unique establishment-level identifier assigned by the LBD, the lbdnum, to track firms over our sample period 1994-2005.

Business establishments in the LBD are linked to ExecuComp and Compustat through the Compustat-SSEL bridge.

C. ExecuComp Data

[7] See Jarmin and Miranda (2002) for more information.
[8] Establishment age is truncated at 30 years in the data.

To estimate the size of broad-based stock option plans, we follow a methodology similar to Oyer and Schaefer (2005). First, we obtain data on the top-five executives' option holdings and option grants from the Standard & Poor (S&P) ExecuComp database. ExecuComp also provides the percentage of total employee option grants assigned to the top-five executives, allowing us to calculate the value of stock options granted to all employees at the firm. Because most firms have executives outside the top five, and we wish to subtract their option grants as well, we assume that non-top five executives receive grants equal to 10% of the average grant awarded to the 2nd through 5th executives. Our estimate of options granted to non-executive employees is, therefore, equal to total grants minus the sum of 1) grants awarded to top 5 executives and 2) ten percent of the average grant made to the top 5 group after subtracting the grant made to the CEO. This variable is restricted to being positive.

We use an indicator variable to identify those firms with the largest BBSO plans. We define a large BBSO plan as occurring at firm *i* at time *t* if the inflation-adjusted Black-Scholes dollar value per employee of new BBSO option grants at firm *i* in year *t* exceeds the 95th percentile of the sample distribution. Given the vesting requirements typically associated with option grants, we assume that a large BBSO grant will impact turnover for three years. As such, the BBSO variable is equal to 1 if the firm had a large BBSO grant this year, last year or the year before, and 0 otherwise.

In defining the BBSO variable, there were three important considerations. First, we need to capture truly broad-based option grants as turnover is being broadly estimated. Therefore, we restrict attention to the largest grants where it becomes increasingly unlikely that such grants could be allocated to just a small group of top executives.

Second, we must make sure that the indicator represents whether employees own unvested options, not simply whether they were issued options in a particular year. Because it is common for options to vest on a three year schedule, we assign a value of 1 to our baseline variable so long as options were granted in the year of the observation, or in the prior two years. As vesting schedules may allow partial vesting prior to three years, we may imagine that the effect of an option grant is decreasing over time. To allow for this, we also sometimes perform regressions with year-specific indicator variables.

Third, we must be sure that our variable is capturing large spikes in option issuance. That is, we must be sure that we are not capturing regular issuances from the same firms over time. Figure 1 shows that roughly 2/5 of firms that have at least one BBSO plan in our data have only one. About 1/4 have two. Figure 2 shows that, of the firms with two BBSO grants in our data, about 2/3 issue the BBSOs in back-to-back years, which essentially amount to one large grant over two years. We are therefore confident that our BBSO indicator variable is capturing grants that are unusual for the firm in question, and broad-based.

D. CRSP and Compustat

We use Compustat to identify firm characteristics and CRSP to measure firm stock performance. We use several control variables that have been suggested in the literature to affect turnover: book leverage, profitability, and size (log of total assets) from Compustat, and firm buy-and-hold stock returns, stock volatility, and sigma from CRSP.

E. Linking data

The business establishments in the LEHD can be linked to the business establishments in the LBD using crosswalk files provided by the US Census. These files identify business establishments in the LEHD which can be matched to business establishments in the LBD based on EIN, industry, state, and county variables. However, two issues arise from matching. First, due to inconsistency in reporting, not all business establishments in the LEHD can be matched to the LBD using EIN, industry, state and county. Second, if a firm has multiple establishments located in the same county and operating in the same line of business, we cannot do a one to one match. Instead, we aggregate the LEHD and LBD data to a "meta-establishment" which includes all of the firm's establishments in that state, county and industry.

We use an iterative process to complete our matching. In each iteration, we identify a match using EIN, state, county and some definition of industry. In the first iteration, we use the most stringent definition of industry, 4-digit SIC (for 1992-2001) or 6-digit NAICS (for 2002-2005). For those establishments not matched in the first iteration, we then attempt a match again using EIN, state and county but now using 3-digit SIC or 4-digit NAICS. In the third iteration we use 2-digit SIC and NAICS codes. In the fourth iteration we use 1-digit SIC or NAICS codes. In the final iteration, we match only on EIN, state and county.

Table 2 presents summary statistics for our dataset and for the sub-sample of firms that have recently implemented a large BBSO plan.[9] Issuing firms are smaller, on average, as measured by assets and employees, but are of more similar market capitalization to firms in the broader sample. They are less levered, and feature considerably higher market-to-book ratios and

[9] Summary statistics in Tables 2 and 3 are created using all ExecuComp data, including observations which cannot be matched to Census data and are dropped from the Tables 4- 10. We report these statistics using the full sample to avoid the more significant disclosure process associated with disclosing means and medians of confidential Census data. Once we are assured of no further changes to the sample, our intention is to clear the results using the same sample as in Tables 4 to 10.

greater stock price volatility. These properties of large BBSO issuers are consistent with those found in the literature. Table 3 reveals that large issuers are found in a wide variety of industries, most commonly business services, pharmaceuticals, and computers and electronic equipment.

II The Effect of BBSO Plans on Turnover

We begin by providing evidence that firms experience a decrease in turnover after issuing stock options to employees. As shown in Table 4, turnover is significantly lower at firms that have recently issued a large BBSO grant to employees. Quarterly turnover in the full sample is 5.7%,[10] while quarterly turnover in the sub-sample of large BBSO issuing firms is 2.9%, approximately half of the level in the full sample.

These data are suggestive that a BBSO can reduce turnover, but are not conclusive. Firms implementing a BBSO plan are, both observably and un-observably, different from firms that do not implement such plans. In Table 5, we attempt to control for cross-sectional and time-series variation in the type of firm implementing a BBSO plan by estimating variants of the following equation:

$$T_{i,j,t} = \alpha BBSO_{i,t-1} + X_{i,j,t-1}\beta + \gamma e_{i,j} + \delta T_{i,j,t-1} + \varepsilon_{i,j,t} \qquad (2)$$

$BBSO_{i,t-1}$ is our indicator for whether a firm has recently implemented a large BBSO plan, $X_{i,j,t-1}$ are macro-, firm-, or establishment-level controls, lagged one period, and $e_{i,j}$ are establishment fixed-effects. We use quarterly establishment-level observations, and cluster our standard errors at the firm level.

[10] Average turnover in our sample is lower than Bureau of Labor Statistics estimates of 9%-10% in the US over the same period, calculated as the ratio of monthly turnover (approximately 4.6 million) to average employment (approximately 140 million). This discrepancy is likely due to the facts that (i) our data only include workers employed at the establishment for at least three months (full quarter employees), and (ii) our data only includes workers at large public firms. We report an employee-weighted average as this facilitated Census disclosure.

In column 1 of Table 5, Panel A, we include the BBSO indicator variable with additional controls and observe a negative and significant coefficient on the BBSO indicator. Thus, the observed reduction in turnover following the adoption of a BBSO plan reported in Table 4 appears to be due to neither a correlation of BBSO plan implementation with time-varying firm characteristics, nor changes in unemployment.

In column 2, we include establishment fixed effects, which control for time-invariant differences at establishments with BBSO plans and those without. We continue to find that turnover at an establishment is reduced following the adoption of a BBSO plan. This result also holds with year fixed effects (column 3).[11] On average, firms experience a 0.48% - 0.52% drop in quarterly turnover in the three years following the implementation of a BBSO plan. Note that this is an absolute change: if a typical firm in this group experiences 5% turnover, then the effect is to reduce the turnover rate by 10%.[12]

These results systematically under-estimate the effect of such plans for two reasons. First, our measure of turnover includes all employees, while it is unlikely that even 50% of employees receive BBSO shares. As such, the effect on employees who receive the treatment (get shares) should be at least twice as large as the overall effect estimated. Second, selection issues may bias our results. For example, perhaps in an attempt to stem a tide of departures, managers adopt BBSO plans when they expect turnover to increase. In this case, the coefficient on our BBSO dummy will underestimate the magnitude of the effect.

Establishment fixed effects control for time-invariant establishment characteristics. An alternative approach is to include lagged turnover, $T_{i,j,t-1}$, as in Carter and Lynch (2004). In column 4, we include lagged turnover but drop establishment fixed effects. As in earlier

[11] In column 3, we drop the control variable, unemployment, because it is an annual variable and is unidentified once we add year fixed effects.
[12] 9.6% = 0.0048/0.05 and 10.4% = 0.0052/0.05.

specifications, we observe a negative and statistically significant coefficient on our BBSO dummy variable.

One might expect that if a firm is doing particularly well, turnover would be lower, as implicit retention incentives – like the higher likelihood of promotion and lower likelihood of layoffs – should be stronger. In column 5, we add the firm's lagged industry-adjusted buy-and-hold returns to our regression to investigate the effect of firm-level success. While the negative coefficient suggests that this intuition is correct, it is not statistically significant.

In Panel B, we investigate the effect of BBSO implementation at an establishment in each year after issuance. BBSOs should reduce turnover while they remain unvested and, potentially, increase turnover in the year following vesting, as employees who postponed a departure to capture value from their options may now be more likely to quit. While we do not observe the vesting period(s) for a given firm's BBSO issue, this period is typically less than or equal to three years. Year 0 is the year of the large BBSO grant. Year 1 is the year after a large BBSO grant. Year 2 (3) captures two (three) years after the large BBSO grant. If all options vest in the first three years (years 0-2 in our definition) then year 3 will reflect the first year where all options are assumed to have vested.

In column 1, we see that while the coefficient in year zero is negative, it is not statistically significant. The more modest effect may be driven by the fact that the options will not vest for several years, making them less important to decisions today. In years one and two, the effect is clearly negative: turnover is reduced in the two years following the year of the BBSO grant. As one would expect if options partially vest in each year, the effect in year three is less than in year two, though the difference is not statistically significant. In the third year after the year of the grant, when most or all of the options are likely to be vested, we see a significant

increase in turnover, consistent with the view that employees time their departures to take place after full vesting. Therefore, managers should expect an increase in turnover after vesting, and plan accordingly.

In columns 2 and 3, we repeat the regression in column 1 but use only a subset of the firms with BBSO plans. For firms which have a BBSO at any point in time, we require that the firm have non-missing information for years 0, 1, 2, and 3. We retain all control firm-year observations used in column 1 and include additional control firm-year observations that were previously dropped from the sample due to missing data for lag turnover.[13] Column 2 includes controls for national unemployment rates. Column 3 includes year fixed effects. We find similar results, indicating that differences in the sample of BBSO firms reflected in each BBSO year dummy was not driving the result in column 1.

The reversal indicates that the majority of the effect of BBSO plans on turnover is simply to delay turnover. We find, on average, that 87% of the reduction in turnover observed in the first three years is reversed in the subsequent year. Furthermore, a number of firms issue multiple large BBSO grants. To the extent that a second large BBSO grant timed years after the first grant reduces future turnover, we are under-estimating the reversal. In unreported results, we repeat the regression in column 2 but exclude firms with multiple BBSO grants and find an even more dramatic reversal.[14]

Across all regressions, three further results are apparent. First, as previously shown by Lane, Isaac, and Stevens (1996) and Porter and Steers (1976), larger firms and older establishments experience lower turnover. Second, as the firm's stock market volatility (sigma) increases, turnover at its establishments increases as well. This fact is robust across

[13] We add these additional control observations as this facilitated disclosure of this sample.
[14] While this result is consistent with an under-estimation of the magnitude of the reversal, it also introduces a look-ahead bias. Firms which do not issue subsequent large BBSO grants may be unique and have higher future turnover.

specifications, and may be because these firms are experiencing significant changes. Third, higher unemployment is associated with lower turnover, as would be expected. These results are largely consistent with prior work.

Together, we have found results consistent with a reduction in turnover resulting from the implementation of a broad based employee stock option plan, but have not shown causality. A variety of stories regarding the choice of the manager to implement a BBSO plan are also consistent with these results. For example, perhaps managers who anticipate that the firm will soon experience a period of success are more likely to share this with employees by adopting large BBSO plans. Because turnover is likely lower during such periods, turnover would be lower following the adoption of a BBSO plan.

We therefore must tease apart different explanations for our results so far. We do so in Sections II.1 and II.2.

II.1 The Effect of BBSO Plans on Turnover in Up vs. Down Markets

In this section, we show that the negative association between the implementation of a BBSO plan and turnover is larger when industry or market returns are high during the post-issuance, pre-vesting period. As argued in Oyer (2004), industry (market) returns reflect labor market conditions: higher industry (market) returns imply better outside employment opportunities for employees with industry-specific (general) human capital. Assuming that a firm's stock price is correlated with those of its peers, the value of previously granted options will be higher when the returns of its peers are high. Thus, options provide a cost to quitting that increases during periods when outside employment options are greatest.

This argument suggests an approach to identify evidence consistent with a causal effect of BBSO plans on employee turnover. Assuming managers cannot predict industry or market returns, realized returns which occur after the options have been issued, but before they have vested, will reflect exogenous shocks to the value of unvested options, and therefore the expected impact of BBSO plans on turnover. Firms with BBSO plans should see greater relative declines in turnover, as compared to firms without BBSO plans, when industry or market returns are higher.[15] Three assumptions are critical to this identification. First, industry/market returns must correlate positively with firm returns. In Section II.2, we use this assumption to further refine the test. Second, industry/market returns must be unpredictable for the manager. This cannot be shown conclusively, but it would be surprising if managers were able to forecast industry/market returns with such a high degree of confidence that they use these expectations to determine the timing of BBSO plans. Third, the correlation of other (dis)incentives for turnover with industry/market returns must be similar for BBSO and non-BBSO issuing firms. That is, one might imagine that firms issuing BBSOs are high Beta. In this case, while a strong market would raise outside opportunities for all firms' employees, the effect would be muted for BBSO issuers because implicit retention incentives would be unusually high. We discuss in Section II.3 below that this does not appear to be a driver of our results.

To test whether decreases in turnover following the implementation of a BBSO plan are indeed greater in magnitude when the market performs well, we use a difference-in-difference type methodology. We interact measures of industry and market performance with our indicator variable, BBSO. If the effect of a BBSO grant is larger in up versus down markets, then the coefficient on this interaction term should be negative. In the absence of one obvious measure of

[15] The relation will be weaker if firms re-price their granted options following weak performance, biasing our results towards zero. However, this is unlikely given that Carter and Lynch (2001) find that firms are unlikely to re-price options in response to poor industry performance.

market performance for our tests, we use three different variants: equal- and value-weighted market returns, and industry median returns. We define industry at the 3-digit SIC code level.

To ensure that the BBSO plan was implemented before the returns were realized, we continue to lag the BBSO indicator variable by one year, as in Table 5. We estimate returns over a nine month window, beginning at the end of the period over which BBSO is measured and ending at the beginning of the period over which turnover is measured. We require the returns to be estimated over a period of time which strictly follows the window for estimating allocated BBSO shares in order to estimate the effect of unanticipated returns at the time of the BBSO issuance. We require the returns to be estimated over a period of time which strictly precedes the window for estimating turnover to avoid reverse causality interpretations if market returns reflect contemporaneous turnover.

Table 6 presents the results of this analysis. In columns 1-3, we find that when the market or industry has performed well, turnover is on average higher, suggesting greater outside employment options for employees. Columns 4-6 include interaction terms. In each case, with statistical significance in columns 5 and 6, the existence of a BBSO plan has a larger effect on subsequent turnover when industry/market returns are high after the fact. This is consistent with a causal connection between BBSO plans and subsequent turnover, and inconsistent with many alternative selection stories.

II.2 The Effect of BBSOs at Firms Whose Performance is More or Less Correlated with Peers

Because we are using industry or market returns as a proxy for the relative value of unvested options, our argument relies on a positive correlation between the performance of the issuing firm and its competitors. We therefore split our sample, in Table 7, between firms whose stock prices are more or less correlated with those of its industry peers.

Column 1 of Table 7 shows that, at firms whose stock prices are not highly correlated with their peers, a BBSO plan does not reduce turnover more when value-weighted market returns are high. However, as shown in column 4, for firms whose stock prices are highly correlated with their peers, BBSO plans are especially effective when the market is performing well. Similar results are shown when we use equal-weighted market returns or industry returns, as shown in columns 2 and 5, and 3 and 6, respectively. The results in Table 7 also hint at significant cross-sectional variation in the effect of BBSOs on turnover, an issue explored in more detail in Section III.

II.3 Alternative Interpretations

We argue that the results in Tables 6 and 7 are consistent with BBSO plans having a causal impact on turnover. However, these same results can also be explained by an omitted variable if firms which adopt BBSO plans have different sensitivities to market returns. For example, if BBSO firms have, on average, stock returns that are more correlated with the market than non-BBSO firms, then they will perform better than their peers in up-markets. This better performance may provide greater career opportunities for employees and result in lower turnover.

To evaluate this alternative interpretation of our results, we follow two approaches. First, we use two proxies for a firm's expected performance in up markets, the historic correlation of a

firm's stock prices with those of the market (Beta), and a measure of growth options (Tobin's Q) at the time of the BBSO grant. We match control firms to our BBSO firms on both of these criteria and then re-run earlier tests using these new samples. Second, we directly test the relation between these proxies and turnover, conditional on market performance. Our earlier results are robust, suggesting that this alterative story is not driving our results.

We estimate a standard market Beta for all of our firms using weekly data over the prior year. Consistent with this alternative hypothesis, we find that BBSO firms have higher Betas as compared to non-BBSO firms. We then create a matched control sample with a similar average Beta to our BBSO firms. Up to three control firms are matched to each BBSO firm in the year of the grant (i.e., year 0). We draw control firms from the sample of all firms in ExecuComp that can be matched to the Census data, but that do not issue a large BBSO grant during our sample period. We require matched firms to operate in the same 3-digit SIC code and in the same year as the BBSO granting firm to which they are being matched. We then pick the three firms with values for Beta closest to the Beta of the BBSO issuer. We allow control firms to match to more than one BBSO issuer.

We repeat the regressions in the last three columns of Table 7 and find the results are robust (unreported)[16]. The coefficient on jump*returns is always still negative and significant. In the regressions using the VW market index or industry returns, the coefficient on the interaction increases in magnitude by approximately 40% and 20%, respectively. In the regression using the EW market index, the coefficient on the interaction is still negative and significant but decreases in magnitude by approximately 10%.

[16] We acknowledge that describing these results without showing the actual regression coefficients is not ideal. Census policies which protect the confidentiality of the data make it more difficult to disclose regression coefficients as compared to text describing the results. We intend to clear regression coefficients in the next major revision.

We repeat this same procedure, but match on Tobin's Q at the time of the BBSO grant. Tobin's Q is calculated as market_equity plus long term debt divided by assets. We again find the results in the last three columns of Table 7 to be robust. In the regressions using the VW market index or industry returns, the coefficient on the interaction term jump*returns is negative and significant and increases in magnitude by approximately 25% and 20%, respectively. In the regression using the EW market index, the coefficient on the interaction is still negative and significant but decreases in magnitude by approximately 10%.

In our second approach, we match on industry, year and firm size (total assets). We then estimate market Beta, Tobin's Q, market-to-book, industry-adjusted market-to-book, and industry market-to-book at the time of the BBSO grant (for BBSO firms), or at the time of the match (for control firms.) We then add each of these year 0 proxies, one at a time, as well as the interaction of each of these proxies with market returns, to the last three regressions in Table 7. This involves 15 new regressions (5 proxies for sensitivity to market returns and 3 definitions of market returns.) In 13 out of the 15 regressions, the interaction of the year 0 proxy and market returns is not negative and significant, indicating that while these variables may predict firm performance conditional on market performance, high values of these variables do not predict an ex-post greater correlation between turnover and market returns.[17] Most importantly, the interaction of BBSO and market returns continues to be negative and significant in 14 out of the 15 specifications.[18]

These facts are difficult to reconcile with non-causal stories associating the implementation of BBSO plans with declines in turnover, and therefore lend credence to the

[17] The two exceptions are the coefficient on the interaction of industry MB and VW returns and on the interaction of industry MB and industry returns are both negative and statistically significant. However, the magnitudes of the two coefficients are small.

[18] The coefficient on the interaction of BBSO and EW returns becomes insignificant (.1 > p > .2) (but remains negative) when market Beta was included as a control and interacted with market returns.

causal argument. They also support Oyer's (2004) argument in favor of deferred equity-based compensation over deferred cash pay.

III The Effect of BBSOs at Market Leading Firms

Having established that BBSOs are effective retention incentives, we now explore how their efficacy depends upon characteristics of the issuing firm. We show that BBSOs are more effective at reducing turnover at market-leading firms, as measured by market-to-book ratio, profit margin, and market share. Each proxy measures a different type of leadership, but for all three, the qualitative effect is the same. We use a matched sample analysis, in which up to three control firms are matched to each BBSO grant firm in the year of the grant (i.e., year 0).[19] We draw control firms from the sample of all firms in ExecuComp that can be matched to the Census data, but that do not issue a large BBSO grant during our sample period. We require matched firms to operate in the same 3-digit SIC code and in the same year as the BBSO granting firm to which they are being matched. We then pick the three firms with values for total assets closest to the total assets of the BBSO issuer. We allow control firms to match to more than one BBSO issuer.

In Table 8, we see that BBSOs are more effective in reducing turnover at high market-to-book firms. Column 1 presents the baseline case, and we see that the effect of BBSO issuance in the matched sample is similar to that found in our prior analysis. In column 2, we interact the BBSO dummy variable with the firm's market-to-book ratio, estimated at the time of the issuance for BBSO granting firms, or in the year of the match for control firms. We find that the

[19] This step reduces our sample size to 842,915 observations.

reduction in turnover is greater when a firm has more growth options.[20] We next test whether growth opportunities relative to peers matter by interacting the BBSO indicator with industry-adjusted market-to-book ratio. As reported in column 3, we find a stronger effect of BBSOs at firms with high growth options relative to their peers.

A high market-to-book ratio could reflect high past returns. If employees over-extrapolate past returns, then high past returns will lead employees to anticipate high future valuations from any new option grants, as in Bergman and Jenter, 2007. As such, a sentiment hypothesis predicts that BBSO grants will have a relatively greater effect on turnover at high market-to-book ratio firms. To further investigate whether a sentiment hypothesis explains our results, we note that employees will care most about total return. As such, a sentiment hypothesis predicts stronger effects of BBSO grants at firms with high industry market-to-book ratios. However, as noted in column 4, we find the opposite result in the data. BBSO grants have a weaker effect on turnover at firms with high industry market-to-book ratios. Finally, if we put industry-adjusted firm market-to-book and industry market-to-book into the same regression, as presented in column 5, the interaction of BBSO with industry-adjusted firm market-to-book is still negative and significant, while industry market-to-book becomes insignificant. It, therefore, appears that BBSOs are more useful at high market-to-book firms for some reason other than sentiment.

In Table 9, we use two alternative measures of market leadership: fraction of sales within a given industry and industry-adjusted profit margin. Column 1 shows that the effect of BBSO plan adoption is greater when the firm has a larger fraction of industry sales, and column 2 shows that the effect is greater when the firm has a higher industry-adjusted profit margin. In each case,

[20] Note, the coefficient on market-to-book is estimated in the regression because one control firm can be matched to multiple BBSO issuers. In that case, market-to-book will be estimated at the year of the match for each unique match. Control firms which are matched to multiple BBSO issuers at different points in time will appear in the data multiple times with different values for the market-to-book ratio for each unique match. This creates within-firm variation in this value in our sample.

the retention incentive provided by a BBSO grant appears to complement incentives provided by the firm's market share and relative profit margin.

Column 3 includes all the three measures we used to proxy for market leadership – industry-adjusted market-to-book, fraction of industry sales, and profit margin – all interacted with the BBSO indicator. Coefficients on all interactions are all negative, with statistical significance for the interaction with industry-adjusted market-to-book and fraction of industry sales. Columns 4 and 5 show that our measures of market leadership are not simply proxies for size, as measured by log assets and log employment. These results, together with the results presented in Table 8, provide evidence that BBSO plans are more effective at market leading firms.

One concern with the above results is that firms which are market leaders may adopt BBSO plans with different time varying selection biases relative to market laggards. We test this alternative interpretation by repeating the regressions in column 4 of Table 6, where we interacted our BBSO indicator variable with market returns, on subsamples of market leading and non-market leading firms.

Our results are reported in Table 10.[21] In columns 1, 3 and 5, we limit the sample to industry leaders, as identified by high industry-adjusted market-to-book, fraction of industry sales or industry-adjusted profit margin. In all three columns we note a negative interaction between the BBSO indicator and value-weighted returns, although the coefficient is not statistically significant in column 5. These results indicate that the greater effect of BBSO plans at market leaders is not entirely driven by selection biases. In columns 2, 4 and 6 we look at

[21] Note that we do not report coefficients on the BBSO indicator variable, nor the number of clusters. This is to satisfy Census disclosure requirements, which restrict reporting of similar results for only marginally differing samples.

industry laggards. We note no evidence of a causal relation between BBSO grants and turnover in the industry laggard sample, consistent with earlier results.

Market leading firms feature lower turnover even in the absence of a BBSO plan, so the fact that the impact of BBSOs on turnover is higher for them is, perhaps, surprising. Our results are suggestive that BBSOs, which are explicit retention incentives, may be complementary to implicit retention incentives, such as a high likelihood of promotion, low likelihood of termination, and higher quality of life at work, which are more present at market leaders. We cannot conclude this with certainty, but we provide suggestive evidence, and see this as an area for future work.

IV Conclusion

Given that the cost of hiring and training a new worker has been estimated to equal between 150% and 175% of her annual pay (Hansen 1997), reducing turnover is a first-order concern for a firm. One potential approach to reduce turnover is to provide the employee with unvested compensation so that she forfeits some wealth if she leaves the firm prior to the vesting date. Unvested stock options may be particularly useful, as they are likely to be worth more when the employee's outside opportunities are most attractive (Oyer, 2004).

We use detailed data to estimate the effect of implementing a broad based stock option (BBSO) plan on employee turnover. Combining quarterly, establishment-level census data on turnover with ExecuComp data on stock option grants, we show that in the years following a large BBSO grant, turnover falls for the granting firm.

We find evidence consistent with causality by exploring exogenous variation in the value of unvested options. If the existence of unvested options is a deterrent to quitting, the deterrent should be larger after a firm's stock price has increased, and the options are more valuable. We cannot look at turnover as a function of firm stock returns directly because they may be predictable for the manager establishing the BBSO plan. Instead, we explore industry and market returns, which occur after the BBSO plan was granted but before the options have vested, and find that pre-existing BBSO plans are more effective at reducing turnover when returns are high.

We document that while BBSO plans are associated with lower turnover during years when the BBSO shares are unvested, this effect is mostly temporary. In the 3rd full year after the year of the BBSO grant, we observe a large increase in turnover suggesting BBSO plans often succeed only in delaying turnover. We find, on average, that 87% of the reduction in turnover observed in the first three years is reversed in the subsequent year. Furthermore, a number of firms issue multiple large BBSO grants. To the extent that a second large BBSO grant timed years after the first grant reduces future turnover, we are under-estimating the reversal.

We also report cross-sectional variation in the effectiveness of these plans. Market leading firms, whether measured by market-to-book ratio, market share, or profit margin, see a larger turnover reduction following the adoption of a BBSO plan. This suggests a complementarity between explicit and implicit retention incentives.

We cannot say that BBSO plans are worth the cost, and we do not suggest that retention is the only justification for employee stock options. They are very expensive, and clearly may serve an incentive role, screen for optimistic employees, etc. Nonetheless, the effect on turnover is also important, and should be considered a first-order benefit of employee stock options.

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Figure 1 – Number of BBSO issuances per firm

This figure presents the distribution of the number of BBSO issuances per firm, among firms with at least one issuance where new BBSO grants exceed the 95th percentile of the sample distribution. Most firms have one or two major broad-based grants in our data.

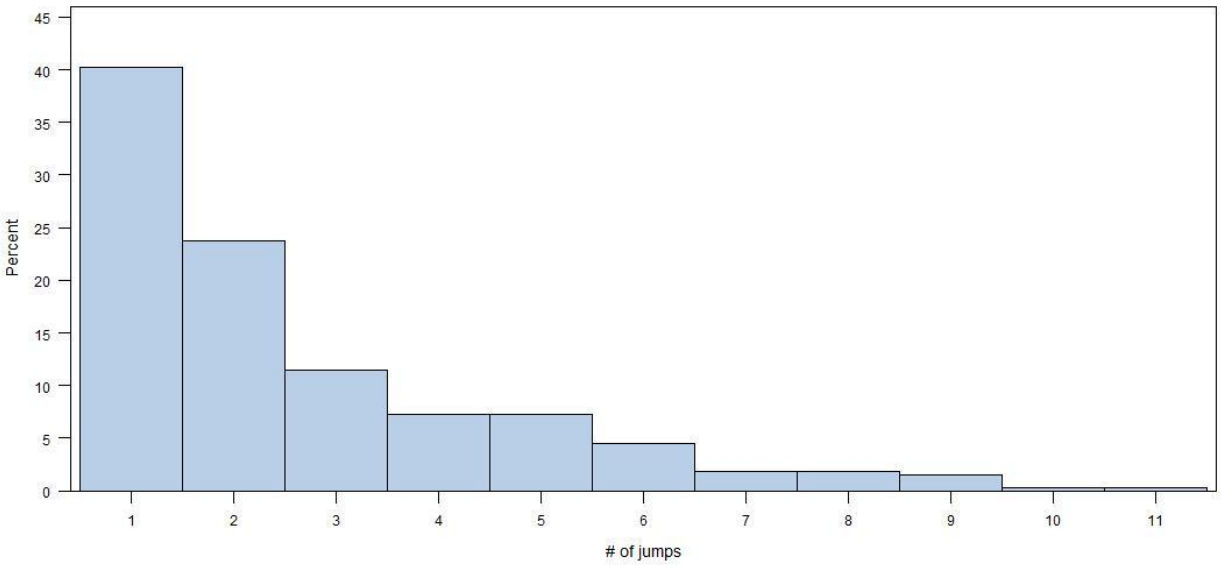


Figure 2 - Number of years between BBSO issuances

This figure presents the distribution for the number of years between BBSO issuances, among firms with exactly two BBSO issuances. In 2/3 of cases, the issuances are back-to-back.

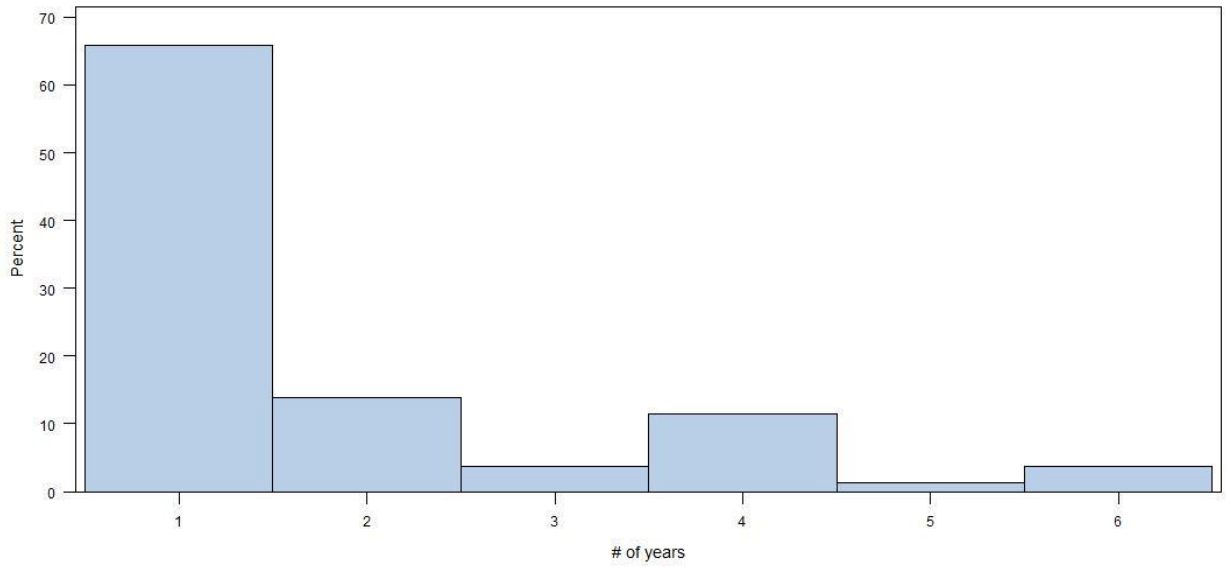


Table 1 - Variable Definitions

Variable	Definition
Turnover	New hires minus change in employment (set to 0 if change in employment is negative) divided by the beginning of period employment at the establishment, measured quarterly for all full-quarter employees
BBSO	An indicator that equals one if the firm's new BBSO option grants per employee (in Black-Scholes dollar value) exceeds the 95th percentile of the sample distribution. In regressions, this variable takes a value of 1 if such a large grant occurred in the current year or in the preceding two years.
Size	Log of total assets, normalized to year 2005 dollars
Market Capitalization	Share price at the end of the fiscal year multiplied by the number of shares outstanding, normalized to year 2005 dollars
Leverage	Long term debt plus debt in current liabilities, divided by assets
Profitability	Ratio of operating income to total assets
One year BNH Returns	Geometric mean of monthly returns for the firm over a 12 month window which corresponds to the firm's fiscal year
BBSO Shares/Employee	Total option grants at a firm minus grants to top-5 executives minus 10% of the average grant to 2nd through 5th executives, divided by the number of employees
BBSO Value/Employee	Black-Scholes dollar value of BBSOs divided by the number of employees
Sigma	Annualized 24-month stock-return volatility
Unemployment	National annual unemployment rate
Establishment Age	Current year minus the first year the establishment appears in the SSEL
Industry Median BNH Returns	9 month buy-and-hold return of median monthly stock return of firms in the same 3-digit SIC. The returns are measured over a window which ends the day before the 3 month period over which turnover is measured
Market-to-book (MB)	Stock price divided by book value per share
Industry-adjusted MB	MB minus average MB of firms in the same 3-digit SIC
Industry MB	Average MB of firms in the same 3-digit SIC
Fraction of Sales	Sales divided by total sales of all public firms in the same 3-digit SIC
Industry-adjusted Profit Margin	Operating income over sales minus average of firms in the same 3-digit SIC

Table 2 – Summary statistics

The following table reports summary statistics for the full sample and the subsample of large BBSO issuers. The full sample includes all firms in ExecuComp for the years 1995 to 2005. The subsample includes all firm-year observations for firms that issued new BBSO option grants per employee (in Black Scholes dollar value) which exceed the 95th percentile of the sample distribution in that year or in the preceding two years. We report medians with means in parentheses. Tests for difference in medians (means) are shown in the right-most columns, where *, **, and *** denote statistical significance at 10%, 5% and 1%, respectively. Variable definitions can be found in Table 1.

Firm-level data	Full Sample	Large BBSO Issuers	Mean Diff Significance	Median Diff Significance
Total assets	1,478.36 (12,332.25)	734.62 (6,282.58)	***	***
Market capitalization	1,446.76 (7,214.08)	1,688.26 (9,843.06)	***	***
Book Leverage	0.21 (0.23)	0.03 (0.14)	***	***
Market-to-book	1.21 (1.81)	2.60 (4.28)	***	***
Profitability	0.13 (0.12)	0.09 (0.07)	***	***
Total employees	5.00 (18.31)	0.88 (4.59)	***	***
One year BNH returns	0.01 (0.01)	0.00 (0.00)	***	***
One year ind. adj. BNH returns	-0.00 (-0.01)	-0.01 (-0.01)	***	***
BBSO shares/employee	78.06 (909.79)	2,688.31(6,628.85)	***	***
BBSO value/employee	726.14 (9,540.84)	36,434.06 (93,084.16)	***	***
Sigma	0.36 (0.43)	0.66 (0.72)	***	***
N	17,345	1,359		

Table 3 – Industry distribution of large BBSO issuers

This table presents the distribution of large BBSO issuers firms among the Fama-French 48 Industries. Large BBSO issuers are defined as ExecuComp firms for the years 1995 to 2005 where the firm issued new BBSO option grants per employee (in Black Scholes dollar value) which exceed the 95th percentile of the sample distribution in that year or in the preceding two years.

Industry Name	***# of firms***
Business Services	88
Electronic Equipment	62
Pharmaceutical Products	40
Computers	40
Petroleum and Natural Gas	18
Trading	16
Communication	14
Insurance	10
Medical Equipment	9
Measuring and Control Equipment	8
Machinery	5
Retail	5
Consumer Goods	4
Business Supplies	2
Banking	2
Healthcare	2
Chemicals	2
Construction	1
Food Products	1
Electrical Equipment	1
Utilities	1
Wholesale	1
Entertainment	1

Table 4 - Average turnover among large BBSO issuers and the full sample

This table reports employee-weighted average quarterly turnover among large BBSO issuers and the full sample. The sample includes all establishments owned by firms in ExecuComp for the years 1995 to 2005 that we were able to match to Census Data and had no missing data on the regression variables.

Establishment-level data	Sample	Large BBSO Issuers	P-value for difference
Turnover mean	0.057	0.029	<0.0001
N of Observations	2,637,053	25,117	

Table 5 – Change in turnover around the adoption of a large BBSO plan

This table presents our baseline OLS estimation results. The sample in Panel A includes all establishments owned by firms in ExecuComp for the years 1995 to 2005 that we were able to match to Census Data and had no missing data on the regression variables. In Panel B, the sample in column 1 includes all observations. In Panel B, columns 2-3, the sample includes a larger set of control firms (we did not require that lag turnover be non-missing) and a smaller set of treatment firms. Firms with a BBSO at any point in time are only included in columns 2-3 if there is non-missing data for BBSO years 0 to 3. The dependent variable is turnover for establishment *i* in quarter *t*. BBSO is an indicator that takes the value of one if that firm's new BBSO option grants per employee (in Black-Scholes dollar value) exceeds the 95th percentile of the sample distribution in that year or in the preceding two years. Variables are defined in Table 1. Standard errors are corrected for firm-level clustering. *, **, and *** denote statistical significance at 10%, 5% and 1%, respectively.

Panel A

	1	2	3	4	5
BBSO	-0.0269*** (0.0100)	-0.0052** (0.0026)	-0.0048** (0.0009)	-0.0136*** (0.0047)	-0.0053** (0.0026)
Book Leverage	-0.0032	0.0061*	0.0055	-0.0014	0.0056
	(0.0097)	(0.0036)	(0.0036)	(0.0046)	(0.0036)
Profitability	0.0447**	-0.0006	0.0004	0.0218**	-0.0001
	(0.0209)	(0.0065)	(0.0067)	(0.0104)	(0.0064)
Sigma	0.0011***	0.0003***	0.0003***	0.0005***	0.0003***
	(0.0003)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Size	-0.0058***	-0.0002	0.0000	-0.0028***	-0.0002
	(0.0009)	(0.007)	(0.0006)	(0.0004)	(0.0007)
Ind. Adj. Firm BNH returns					-0.0056
					(0063)
Log Age	-0.0032***	-0.0045***	-0.0035***	-0.0018***	-0.0045***
	(0.0008)	(0.0008)	(0.0009)	(0.0004)	(0.0008)
Unemployment	-0.0032***	-0.0048***		-0.0016***	-0.0048***
	(0.0011)	(0.0006)		(0.0005)	(0.0006)
Lagged Turnover				0.5238***	
				(0.0166)	
Establishment FE	No	Yes	Yes	No	Yes
Year FE	No	No	Yes	No	No
N clusters	1,787	1,787	1,787	1,787	1,787
N	2,637,053	2,637,053	2,637,053	2,637,053	2,637,053
R-square	0.0972	0.5611	0.5614	0.3431	0.5611

Panel B

	1	2	3
BBSO year 0	-0.0009	-0.0011	-0.0011
	(0.0009)	(0.0015)	(0.0013)
BBSO year 1	-0.0044***	-0.0041	-0.0046*
	(0.0010)	(0.0026)	(0.0027)
BBSO year 2	-0.0033**	-0.0040***	-0.0039***
	(0.0014)	(0.0014)	(0.0013)
BBSO year 3	0.0075**	0.0077**	0.0073**
	(0.0034)	(0.0038)	(0.0036)
Book Leverage	0.0061	0.0088 **	0.0084**
	(0.0036)	(0.0039)	(0.0040)
Profitability	-0.0004	0.0008	0.0022
	(0.0065)	(0.0062)	(0.0064)
Sigma	0.0003***	0.0004***	0.0005***
	(0.0001)	(0.0001)	(0.0001)
Size	-0.0002	-0.0002	-0.0002
	(0.0007)	(0.0007)	(0.0007)
Log Age	-0.0045**	-0.0021**	-0.0010
	(0.0008)	(0.0009)	(0.0010)
Unemployment	-0.0048***	-0.0045***	
	(0.0006)	(0.0005)	
Establishment FE	Yes	Yes	Yes
Year FE	No	No	Yes
N clusters	1,787	1,824	1,824
N	2,637,053	3,087,308	3,087,308
R-square	0.5611	0.5384	0.5386

Table 6 – The effect of BBSOs on turnover in up- versus down-markets

The dependant variable is turnover for establishment *i* in quarter *t*. The sample includes all establishments owned by firms in ExecuComp for the years 1995 to 2005 that we were able to match to Census Data and had no missing data on the regression variables. BBSO is an indicator that takes the value of one if that firm's new BBSO option grants per employee (in Black-Scholes dollar value) exceeds the 95th percentile of the sample distribution in that year or in the preceding two years. Variables are defined in Table 1. Standard errors are corrected for firm-level clustering. *, **, and *** denote statistical significance at 10%, 5% and 1%, respectively.

	1	2	3	4	5	6
BBSO	-0.0049**	-0.0050**	-0.0047*	-0.0050**	-0.0047**	-0.0061**
	(0.0022)	(0.0022)	(0.0028)	(0.0022)	(0.0022)	(0.0026)
Value Weighted Market BNH Return	0.0049***			0.0049***		
	(0.0014)			(0.0014)		
BBSO * Value Weighted Market BNH Return				-0.0081		
				(0.0053)		
Equal Weighted Market BNH Return		0.0034***			0.0034***	
		(0.0009)			(0.0009)	
BBSO * Equal Weighted Market BNH Return					-0.0057*	
					(0.0033)	
Industry Median BNH Return			0.0034**			0.0036**
			(0.0017)			(0.0018)
BBSO * Industry Median BNH Return						-0.0075*
						(0.0042)
Book Leverage	0.0096***	0.0090***	0.0062*	0.0096***	0.0090***	0.0062*
	(0.0035)	(0.0035)	(0.0035)	(0.0035)	(0.0035)	(0.0036)
Profitability	0.0095	0.0091	-0.0011	0.0096	0.0091	-0.0011
	(0.0072)	(0.0073)	(0.0067)	(0.0072)	(0.0073)	(0.0067)
Sigma	0.0006***	0.0006***	0.0003***	0.0006***	0.0006***	0.0003***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Size	-0.0000	-0.0002	-0.0001	-0.0000	-0.0002	-0.0001
	(0.0006)	(0.0006)	(0.0007)	(0.0006)	(0.0006)	(0.0007)
Log Age	-0.0067***	-0.0074***	-0.0043***	-0.0067***	-0.0074***	-0.0043***
	(0.0009)	(0.0009)	(0.0008)	(0.0009)	(0.0009)	(0.0008)
Unemployment			-0.0049***			-0.0049***
			(0.0006)			(0.0006)
Establishment fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes
N of clusters	1,713	1,713	1,713	1,713	1,713	1,713
N of observations	2,252,857	2,252,857	2,252,857	2,252,857	2,252,857	2,252,857
R-squared	0.5829	0.5829	0.5831	0.5829	0.5829	0.5831

Table 7 – The effect of BBSOs on turnover in up- versus down-markets, depending upon the correlation of the firm's stock price with the prices of peers

This table presents results on two sub-samples segmented by the correlation in monthly firm and industry returns. The high correlation sub-sample includes all firms with a correlation above the 50th percentile of the correlation distribution. The low correlation subsample includes all the firms with a correlation below the 50th percentile of the correlation distribution. The dependent variable is turnover for establishment *i* in quarter *t*. BBSO is an indicator that takes the value of one if that firm's new BBSO option grants per employee (in Black-Scholes dollar value) exceeds the 95th percentile of the sample distribution in that year or in the preceding two years. Variables are defined in Table 1. Standard errors are corrected for firm-level clustering. *, **, and *** denote statistical significance at 10%, 5% and 1%, respectively.

	1	2	3	4	5	6
Sample	Low Correlation	Low Correlation	Low Correlation	High Correlation	High Correlation	High Correlation
BBSO	0.0018	0.0009	-0.0010	-0.0074***	-0.0064**	-0.0083***
	(0.0021)	(0.0021)	(0.0021)	(0.0026)	(0.0027)	(0.0028)
Value Weighted Market BNH Return	0.0041**			0.0046*		
	(0.0021)			(0.0027)		
BBSO * Value Weighted Market BNH Return	-0.0044			-0.0113**		
	(0.0103)			(0.0052)		
Equal Weighted Market BNH Return		0.0020			0.0043***	
		(0.0015)			(0.0016)	
BBSO * Equal Weighted Market BNH Return		0.0026			-0.0136***	
		(0.0063)			(0.0048)	
Industry Median BNH Return			0.0028			0.0038
			(0.0025)			(0.0032)
BBSO * Industry Median BNH Return			0.0017			-0.0139**
			(0.0048)			(0.0068)
Book Leverage	0.0045	0.0037	0.0008	0.0118	0.0121*	0.0100
	(0.0043)	(0.0043)	(0.0036)	(0.0073)	(0.0072)	(0.0071)
Profitability	-0.0039	-0.0058	-0.0164*	0.0276**	0.0282**	0.0155
	(0.0114)	(0.0115)	(0.0086)	(0.0126)	(0.0125)	(0.0126)
Sigma	0.0005***	0.0005***	0.0001	0.0009	-0.0003	-0.0018
	(0.0001)	(0.0001)	(0.0001)	(0.0048)	(0.0046)	(0.0044)
Size	0.0004	0.0002	0.0005	-0.0014	-0.0016	-0.0017
	(0.0008)	(0.0008)	(0.0008)	(0.0013)	(0.0013)	(0.0014)
Log Age	-0.0090***	-0.0096***	-0.0058***	-0.0039***	-0.0044***	-0.0017
	(0.0016)	(0.0015)	(0.0013)	(0.0014)	(0.0015)	(0.0015)
Unemployment			-0.0050***			-0.0052***
			(0.0008)			(0.0009)
Establishment fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes
N of clusters	1,525	1,525	1,525	1,553	1,553	1,553
N of observations	1,314,083	1,314,083	1,314,083	1,322,970	1,322,970	1,322,970
R-Squared	0.5985	0.5984	0.6009	0.5822	0.5823	0.5843

Table 8 – The effect of BBSO plans on employee turnover by firm value

This table presents the OLS estimation results of the main model on the matched sample with additional interaction terms to explore the change in the effectiveness of BBSO plans at firms with higher and lower market-to-book, measured at the time of BBSO issuance (for issuing firms) and the year they were matched (for control firms). The matched sample is created by matching three control firms that operate in the same 3-digit SIC, are closest in size, and never issue a large BBSO grant, taken from the sample of all firms in ExecuComp. The dependant variable is turnover for establishment *i* in quarter *t*. BBSO is an indicator that takes the value of one if the firm's new BBSO option grants per employee (in Black-Scholes dollar value) exceeds the 95th percentile of the sample distribution in that year or in the preceding two years. Variables are defined in Table 1. Standard errors are corrected for firm-level clustering. *, **, and *** denote statistical significance at 10%, 5% and 1%, respectively.

	1	2	3	4	5
BBSO	-0.0060**	-0.0053*	-0.0059**	-0.0077**	-0.0065**
	(0.0031)	(0.0031)	(0.0029)	(0.0034)	(0.0031)
Market-to-Book		0.0000			
		(0.0000)			
BBSO * Market-to-Book		-0.0002**			
		(0.0001)			
Industry-adjusted Market-to-Book			0.0000*		0.0000
			(0.0000)		(0.0000)
BBSO * Industry-adjusted Market-to-Book			-0.0006***		-0.0006***
			(0.0002)		(0.0002)
Industry Market-to-Book				-0.0000	0.0000
				(0.0000)	(0.0000)
BBSO * Industry Market-to-Book				0.0004**	0.0001
				(0.0002)	(0.0001)
Book Leverage	0.0009	0.0008	0.0008	0.0009	0.0008
	(0.0044)	(0.0044)	(0.0044)	(0.0045)	(0.0044)
Profitability	-0.0056	-0.0055	-0.0055	-0.0057	-0.0055
	(0.0077)	(0.0076)	(0.0076)	(0.0076)	(0.0076)
Sigma	-0.0071***	-0.0071***	-0.0071***	-0.0072***	-0.0071***
	(0.0021)	(0.0021)	(0.0021)	(0.0021)	(0.0021)
Size	-0.0019	-0.0019	-0.0020	-0.0020	-0.0020
	(0.0015)	(0.0014)	(0.0014)	(0.0014)	(0.0014)
Log Age	-0.0034***	-0.0034***	-0.0034***	-0.0034***	-0.0034***
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)
Unemployment	-0.0057***	-0.0057***	-0.0057***	-0.0057***	-0.0057***
	(0.0007)	(0.0007)	(0.0007)	(0.0007)	(0.0007)
Establishment fixed effects	Yes	Yes	Yes	Yes	Yes
N of clusters	522	522	522	522	522
N of observations	842,915	842,915	842,915	842,915	842,915
R-squared	0.6358	0.6358	0.6358	0.6358	0.6358

Table 9 – The effect of BBSO plans on employee turnover by market leadership

This table presents the OLS estimation results of the main model on the matched sample with additional interaction terms to explore the effectiveness of BBSO plans as a function of firm market share, profit margin, market-to-book, and size, all measured at the time of BBSO issuance (for issuing firms) and the year they were matched (for control firms). The matched sample is created by matching three control firms that operate in the same 3-digit SIC, are closest in size, and never issue a large BBSO grant, taken from the sample of all firms in ExecuComp. The dependant variable is turnover for establishment *i* in quarter *t*. BBSO is an indicator that takes the value of one if that firm's new BBSO option grants per employee (in Black-Scholes dollar value) exceeds the 95th percentile of the sample distribution in that year or in the preceding two years. Variables are defined in Table 1. Standard errors are corrected for firm-level clustering. *, **, and *** denote statistical significance at 10%, 5% and 1%, respectively.

	1	2	3	4	5
BBSO	-0.0060**	-0.0053*	-0.0015	-0.0059**	-0.0077**
	(0.0031)	(0.0031)	(0.0017)	(0.0029)	(0.0034)
Fraction of Ind. Sales	0.0002**		0.0002*		
	(0.0001)		(0.0001)		
BBSO * Fraction of Ind. Sales	-0.0003***		-0.0003***		
	(0.0001)		(0.0001)		
Ind. Adj. Profit Margin		0.0012	-0.0008		
		(0.0011)	(0.0019)		
BBSO * Ind. Adj. Profit Margin		-0.0168***	-0.0118		
		(0.0060)	(0.0074)		
Ind. Adj. MB			0.0000*		
			(0.0000)		
BBSO * Ind. Adj. MB			-0.0001*		
			(0.0000)		
Log Size				0.0010**	
				(0.0004)	
BBSO * Log Size				-0.0003	
				(0.0009)	
Log Employment					0.0009***
					(0.0004)
BBSO * Log Employment					-0.0004
					(0.0013)
Book Leverage	0.0009	0.0007	0.0008	0.0008	0.0005
	(0.0045)	(0.0044)	(0.0045)	(0.0045)	(0.0045)
Profitability	-0.0050	-0.0058	-0.0051	-0.0051	-0.0062
	(0.0077)	(0.0076)	(0.0077)	(0.0077)	(0.0078)
Sigma	-0.0074***	-0.0070***	-0.0070***	-0.0070***	-0.0070***
	(0.0021)	(0.0021)	(0.0021)	(0.0021)	(0.0021)
Size	-0.0019	-0.0020	-0.0020	-0.0020	-0.0022
	(0.0015)	(0.0014)	(0.0015)	(0.0015)	**(**0.0015**)**
Log Age	-0.0034***	-0.0034***	-0.0034***	-0.0034***	-0.0034***
	(0.0012)	(0.0011)	(0.0012)	(0.0012)	(0.0012)
Unemployment	-0.0058***	-0.0057***	-0.0058***	-0.0058***	-0.0057***
	(0.0006)	(0.0007)	(0.0006)	(0.0006)	(0.0007)
Establishment fixed effects	Yes	Yes	Yes	Yes	Yes
N of clusters	522	522	522	522	522
N of observations	842,915	842,915	842,915	842,915	842,915
R-squared	0.6359	0.6358	0.6358	0.6358	0.6358

Table 10 – The effect of BBSOs on turnover in up versus down markets by market leadership

This table estimation results on sub-samples divided by measures of market leadership at the time of the issuance(for issuing firms) and the year they were matched (for control firms). The top (bottom) sub-samples include all firms above the 75th percentile (below the 25th percentile) of the market-to-book, fraction of industry sales and profit margin distributions. Market performance is measured by value weighted market buy-and-hold returns. The dependent variable is turnover for establishment *i* in quarter *t*. BBSO is an indicator that takes the value of one if that firm's new BBSO option grants per employee (in Black-Scholes dollar value) exceeds the 95th percentile of the sample distribution in that year or in the preceding two years. All variables are defined in Table 1. Standard errors are corrected for firm-level clustering. *, **, and *** denote statistical significance at 10%, 5% and 1%, respectively. The coefficient on the BBSO indicator and number of clusters are suppressed in all columns due to Census disclosure requirements.

	1	2	3	4	5	6
Sample	Top Ind. Adj. M/B	Bottom Ind. Adj. M/B	Top Frac. Ind. Sales	Bottom Frac. Ind. Sales	Top Ind. Adj. Margin	Bottom Ind. Adj. Margin
BBSO	XXX	XXX	XXX	XXX	XXX	XXX
Value Weighted Market	0.0052**	0.0042**	0.0093**	0.0033	0.0097	0.0053**
BNH Return	(0.0025)	(0.0018)	(0.0044)	(0.0029)	(0.0070)	(0.0025)
BBSO * Value Weighted	-0.0130*	0.0027	-0.0094*	-0.0057	-0.0173	-0.0026
Market BNH Return	(0.0069)	(0.0048)	(0.0054)	(0.0093)	(0.0124)	(0.0045)
	0.0100**	-0.0050	-0.0084	-0.0032	0.0097	-0.0049
Book Leverage	(0.0048)	(0.0068)	(0.0078)	(0.0126)	(0.0126)	(0.0060)
	0.0065	0.0053	-0.0115	0.0046	0.0317*	0.0050
Profitability	(0.0081)	(0.0098)	(0.0110)	(0.0116)	(0.0189)	(0.0093)
	-0.0256***	-0.0077**	-0.0107**	-0.0144**	0.0051	-0.0051
Sigma	(0.0043)	(0.0032)	(0.0052)	(0.0070)	(0.0083)	(0.0041)
	-0.0013	-0.0027	-0.0031	-0.0047	0.0057	-0.0031
Size	(0.0019)	(0.0028)	(0.0035)	(0.0052)	(0.0040)	(0.0022)
	-0.0090***	-0.0063***	-0.0075***	-0.0054	-0.0061	-0.0054***
Log Age	(0.0016)	(0.0017)	(0.0023)	(0.0051)	(0.0038)	(0.0093)
Establishment fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N of clusters	XXX	XXX	XXX	XXX	XXX	XXX
N of observations	49,449	475,830	264,704	104,631	86,281	468,441
R-squared	0.5083	0.6737	0.7364	0.4225	0.4091	0.6732